

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

First Order Logic

Outline

- First Order Logic
- Prolog

FIRST ORDER LOGIC (FOL)

First Order Logic

- A major problem with propositional logic is that entire propositions are represented as a single symbol.
- For example, the sentence “Ali likes cricket” is represented as a symbol, say AC. But we can’t represent Ali, likes, and cricket as separate entities. Also, we can’t represent how they are connected.
- FOL is a representation that permits finer control and more accurate representation of the environment.
- It is also referred to as Predicate Calculus.
- Using FOL, we can add greater expressiveness as well as more generalization to our knowledge.

First Order Logic

- Propositional logic assumes the world contains facts (true or false).
- First-order logic (like natural language) assumes the world contains
 - **Objects:**
 - people, houses, numbers, colors, baseball games, wars, ...
 - **Relations:**
 - red, round, prime, brother of, bigger than, part of, comes between, ...
 - **Functions:**
 - father of, best friend, one more than, plus, ...

First Order Logic – Syntax

- In First Order Logic knowledge is built up from
- **Constants** (the objects of the knowledge),
- A set of **predicates** (relationships between the objects of knowledge), and
- Some number of **functions** (indirect references to other knowledge)
- In FOL, we use three kinds of symbols: Object symbols, predicate symbols, and function symbols.

First Order Logic – Atomic Sentences

- Constant refers to a single object
 - Marc, Elise, Bicycle, Scooter, the-Stranger, Colorado
- Predicates are used to define relationships and properties of constants
 - Person(Marc) Marc is a Person
 - Person(Elise) Elise is a Person
- Note that predicates have arity, which refers to the number of arguments.
- The 'Person' predicate has an arity of one where the predicate 'knows' has an arity of two.

First Order Logic – Atomic Sentences

- For example,
 - `Knows(Marc, Elise)` Marc and Elise knows each other
 - `Rides(Marc, Bicycle)` Marc rides a Bicycle
 - `Owns(Elise, Book(the-stranger))` Elise owns a book called the-stranger
- A function allows us to transform a constant into another constant. For example, the `sister_of` function is demonstrated below:
 - `Knows( marc, sister_of( sean ) )` Marc knows Sean's sister.
- Functions are relations in which there is only one “value” for a given “input.”

First Order Logic – Compound Sentences

- Two or more atomic sentences can be used to build a compound sentence using connectives
 - And ($\wedge$)
 - Or ($\vee$)
 - Not ($!$)
 - Logical Conditional ($\Rightarrow$)
 - Logical Bi-conditional/Equivalence ($\Leftrightarrow$)
- Marc and Elise know one another
 - $\text{Knows}(\text{Marc}, \text{Elise}) \wedge \text{Knows}(\text{Elise}, \text{Marc})$
- If Marc knows Elise, then Elise knows Marc
 - $\text{Knows}(\text{Marc}, \text{Elise}) \Rightarrow \text{Knows}(\text{Elise}, \text{Marc})$

First Order Logic – Compound Sentences

- Marc knows Elise, and Elise does not know Marc
- Marc rides a scooter or Marc rides a bicycle
- If Marc knows Elise, then Elise knows Marc

First Order Logic – Variables

- Most of the time not all the information is present
- Variables help to construct abstract sentences that do not specify specific objects
- ***If x knows Elise, then x is a Person***
- ***$\text{Knows}(x, \text{Elise}) \Rightarrow \text{Person}(x)$***
- If we know that: $\text{Knows}(\text{Marc}, \text{Elise})$ then we could deduce that Marc is a Person

First Order Logic - Quantifiers

- Quantifier – used to determine the quantity of a variable
- Two quantifiers in FOL
- ***Universal Quantifier*** ($\forall$) (for all, everyone, everything)
 - Used to indicate that a sentence should hold when anything is substituted for the variable.
 - We use implication " $\rightarrow$ " symbol.
 - Example:
 $\forall x \text{ fish}(x) \rightarrow \text{live}(x, \text{water}).$
- Read as: There are all x where x is a fish who lives in water.

First Order Logic - Quantifiers

- Quantifier – used to determine the quantity of a variable
- Two quantifiers in FOL
- ***Existential Quantifier*** ($\exists$) (for some, at least one)
 - Indicates that there is something that can be substituted for the variable such that the sentence holds
 - We always use AND or Conjunction symbol ($\wedge$).
 - Example:
$$\exists x: \text{prime}(x) \wedge \text{even}(x)$$
 - There are some x where x is a prime which is even.

First Order Logic - Quantifiers

- $\forall x \text{ MAN}(x)$
- $\forall x \text{ CHASE}(x,s)$
- $\forall x \text{ WANT}(m,x)$
- $\forall x \forall y \text{ LOVE}(x,y)$

Everything is a man

Everything chased Socrates

Mario wanted everything

Everything loves everything

- $\exists x \text{ MAN}(x)$
- $\exists x \text{ CHASE}(x,s)$
- $\exists x \text{ WANT}(m,x)$
- $\exists x \exists y \text{ LOVE}(x,y)$

Something is a man

Something chased Socrates

Mario wanted something

Something loves something

First Order Logic - Quantifiers

- Brothers are siblings

$$\forall x,y \text{ Brother}(x,y) \Leftrightarrow \text{Sibling}(x,y)$$

- One's mother is one's female parent

$$\forall m,c \text{ Mother}(m) \Leftrightarrow (\text{Female}(m) \wedge \text{Parent}(m,c))$$

- “Sibling” is symmetric

$$\forall x,y \text{ Sibling}(x,y) \Leftrightarrow \text{Sibling}(y,x)$$

- Every gardener likes the sun.

$$\forall x \text{ gardener}(x) \Rightarrow \text{likes}(x, \text{Sun})$$

First Order Logic and Prolog

- Prolog – Programming in Logic
- Declarative language
- Language for the representation of knowledge
- Allows the definition of facts (clauses) and rules (clause with body)

First Order Logic and Prolog

- Computer programming in Prolog consists of:
 1. Specifying some ***facts*** about objects and their relationships
 2. Defining some ***rules*** about objects and their relationships
 3. Asking questions about objects and their relationships

First Order Logic and Prolog

- Prolog program is a database of facts and rules
- Some are always true – facts
 - `father( john, jim).`
- Some are dependent on others being true – rules
 - `parent( Person1, Person2 ) :- father( Person1, Person2 );`
 - `mother( Person1, Person2 ).`
- To run a program, we ask questions about the database

First Order Logic and Prolog - Predicate

- 'Predicate' is the name given to the word occurring before the bracket in a fact or rule:
 - `parent(jane,alan).`
- By defining a predicate, you are specifying which information needs to be known for the property denoted by the predicate to be true.
- Both facts and rules are predicate definitions.

First Order Logic and Prolog - Clause

- A clause consists of a head and sometimes a body.
- `mother(jane,alan).`
 - (Fact)
 - It is a clause.
- `parent(P1,P2):- mother(P1,P2).`
 - (Rule)
 - It is a clause.
 - Here `mother(P1,P2).` is the body of the clause
- Facts don't have a body because they are always true.

First Order Logic and Prolog - Clause

- A predicate head consists of a predicate name and sometimes some arguments contained within brackets and separated by commas.
 - `mother(jane,alan).`
- Arguments consist of terms, which can be:
 - Constants e.g. `jane`,
 - Variables e.g. `Person1`, or `Person 2`
- A body can be made up of any number of subgoals (calls to other predicates) and terms.

First Order Logic and Prolog - Example

- Example Database:
- **Facts:**
 - John is the father of Jim.
 - Jane is the mother of Jim.
 - Jack is the father of John.
- **Rules:**
 - Person 1 is a parent of Person 2 **if**
 - Person 1 is the father of Person 2 **or**
 - Person 1 is the mother of Person 2.
 - Person 1 is a grandparent of Person 2 **if**
 - some Person 3 is a parent of Person 2 **and**
 - Person 1 is a parent of Person 3.

First Order Logic and Prolog - Example

- Example Questions
 1. Who is Jim's father?
 2. Is Jane the mother of Fred?
 3. Is Jane the mother of Jim?
 4. Does Jack have a grandchild?

Example Database:

John is the father of Jim.
Jane is the mother of Jim.
Jack is the father of John.

Person 1 is a parent of Person 2 if
Person 1 is the father of Person 2 or
Person 1 is the mother of Person 2.

Person 1 is a grandparent of Person 2 if
some Person 3 is a parent of Person 2 and
Person 1 is a parent of Person 3.

Example questions:

Who is Jim's father?
Is Jane the mother of Fred?
Is Jane the mother of Jim?
Does Jack have a grandchild?

Example Database:

father(john, jim).
mother(jane, jim).
father(jack, john).

parent(Person1, Person2) :-
father(Person1, Person2).
parent(Person1, Person2) :-
mother(Person1, Person2).

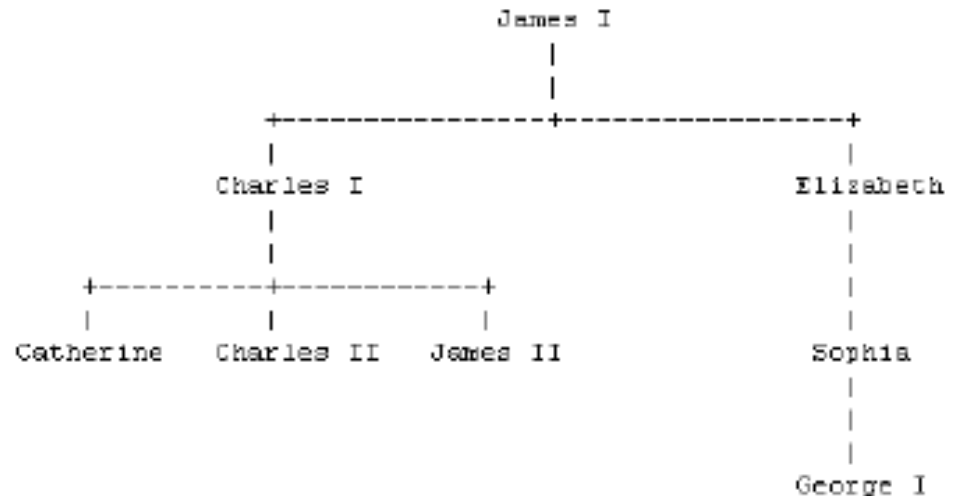
grandparent(Person1, Person2) :-
parent(Person3, Person2),
parent(Person1, Person3).

Example questions:

?- **father(Who, jim).**
?- **mother(jane, fred).**
?- **mother(jane, jim).**
?- **grandparent(jack, _).**

Prolog – Practice Questions

```
male(james1). male(charles1).
male(charles2). male(james2).
male(george1). female(catherine).
female(elizabeth). female(sophia).
parent(james1, charles1).
parent(james1, elizabeth).
parent(charles1, charles2).
parent(charles1, catherine).
parent(charles1, james2).
parent(elizabeth, sophia).
parent(sophia, george1).
```



Write Prolog statements to answer the following queries

Was George I the parent of Charles I?

Who was Charles I's parent?

Who were the children of Charles I?

M is the mother of X if she is a parent of X and is female

F is the father of X if he is a parent of X and is male

X is a sibling of Y if they both have the same parent.

Prolog – Practice Questions

- Was George I the parent of Charles I?
 - `parent(geroge1, charles1).`
- Who was Charles I's parent?
 - `parent(X, charles1).`
- Who were the children of Charles I?
 - `parent(charles1, X).`
- M is the mother of X if she is a parent of X and is female
 - `mother(M,X) :- female(M), parent(M,X).`
- F is the father of X if he is a parent of X and is male
 - `father(F,X) :- male(F), parent(F,X).`
- X is a sibling of Y if they both have the same parent.
 - `sibling(X,Y) :- parent(Z,X), parent(Z,Y), X \= Y.`
 - `X \= Y` is equivalent to “X not equals to Y”.

Prolog – Practice Questions

- Facts about planets
 - `orbits(mercury, sun).`
 - `orbits(venus, sun).`
 - `orbits(earth, sun).`
 - `orbits(mars, sun).`
- Facts about moons
 - `orbits(moon, earth).`
 - `orbits(phobos, mars).`
 - `orbits(deimos, mars).`
- An object P is a planet if it orbits sun.
- An object S is a satellite if it orbits a planet P

Prolog – Practice Questions

- Facts:

- `prereq(introCS,dataStructs).`
- `prereq(dataStructs,databases).`
- `prereq(dataStructs,graphics).`
- `prereq(linAlg,graphics).`

- Rules:

- A course R is a required course for course C if R is a prerequisite of C.
 - `required(Course,NextCourse) :- prereq(Course,NextCourse).`
- A course R is a required course for course C if R is a prerequisite of some course S which is a required course for C.
 - `required(Course,NextCourse) :-
prereq(Course,SomeOtherCourse),required(SomeOtherCourse,NextCourse).`

Prolog – Practice Questions

- What would be result of following queries
 - ? prereq(X,graphics)
 - ? prereq(dataStructures,X)
 - ? Prereq(X,Y)
 - ? required(X,graphics),not(required(X,databases)).

Prolog – Practice Questions

- Suppose someone has already written Prolog clauses that define the following relationships:
 - `male(X)`
 - `female(X)`
 - `parent(X, Y)`
- Write Prolog clauses to define the following relationships:
 - `is_mother(X)`, `is_father(X)`, `is_son(X)`, `sister_of(X, Y)`, `grandpa_of(X, Y)`, `sibling(X, Y)`
- Example
 - `Aunt(X, Y) :- sister_of(X, Z), parent(Z, Y)`

Exercise 1

- ***Facts***

- Burger is a food
- Sandwich is a food
- Pizza is a food
- Sandwich is a lunch
- Pizza is a dinner

- ***Rules***

- Anything is a meal if it is a food

- ***Queries / Goals***

- Is pizza a food?
- Which food is meal and lunch?
- Is sandwich a dinner?

Exercise 2

- ***Facts***

- charlie studies csc135, olivia studies csc135
- jack studies csc131, arthur studies csc134
- kirke teaches csc135, collins teaches csc131
- collins teaches csc171, juniper teaches csc134

- ***Rules***

- X is a professor of Y if X teaches C and Y studies C

- ***Queries / Goals***

- What does charlie study?
- Who are the students of professor kirke?

Exercise 3

- ***Facts***

- jack owns bmw car
- john owns chevy car
- olivia owns civic car
- jane owns chevy car
- bmw car is sedan
- civic car is sedan
- chevy car is truck

- ***Queries / Goals***

- What does john own?
- Does john own something?
- Who owns car chevy?
- Does jane own sedan?